

PrintSensei

An Intelligent Multimodal Robotic Labeling and Knowledge Assistant

1. Project Topic

PrintSensei: An Intelligent Multimodal Robotic Labeling and Knowledge Assistant

2. Project Description

PrintSensei is a Raspberry Pi-based intelligent robotic assistant that accepts information through voice commands, camera images, printed documents, and physical objects. It processes these inputs using speech recognition, computer vision, optical character recognition, object detection, local AI models, and cloud-based AI APIs.

After understanding the user's request, the system automatically generates a suitable label, study card, inventory sticker, product tag, or QR-enabled knowledge label and prints it using a 58 mm thermal printer.

The system is designed as a stationary desktop robot rather than a moving robot. Its main physical action is the automatic generation and printing of useful labels.

Example user commands

- "Create a label for this Arduino Uno."
 - "Summarize this textbook page."
 - "Print a QR code for my website."
 - "Create an inventory sticker for this component."
 - "Make a revision card for binary search."
 - "Read the expiry date and create a product label."
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3. Problem Statement

Creating labels, study notes, inventory tags, QR stickers, and product labels normally requires users to enter information manually, choose a template, format the content, and operate a printer.

Conventional label printers cannot:

- Understand natural-language commands
- Recognize objects
- Read text from documents
- Generate summaries
- Retrieve knowledge
- Decide the most suitable label format

PrintSensei solves this problem by combining multimodal artificial intelligence with a physical printing system.

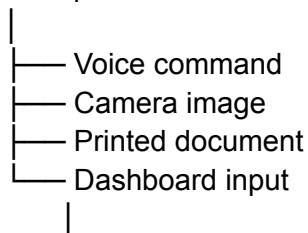
4. Proposed Solution

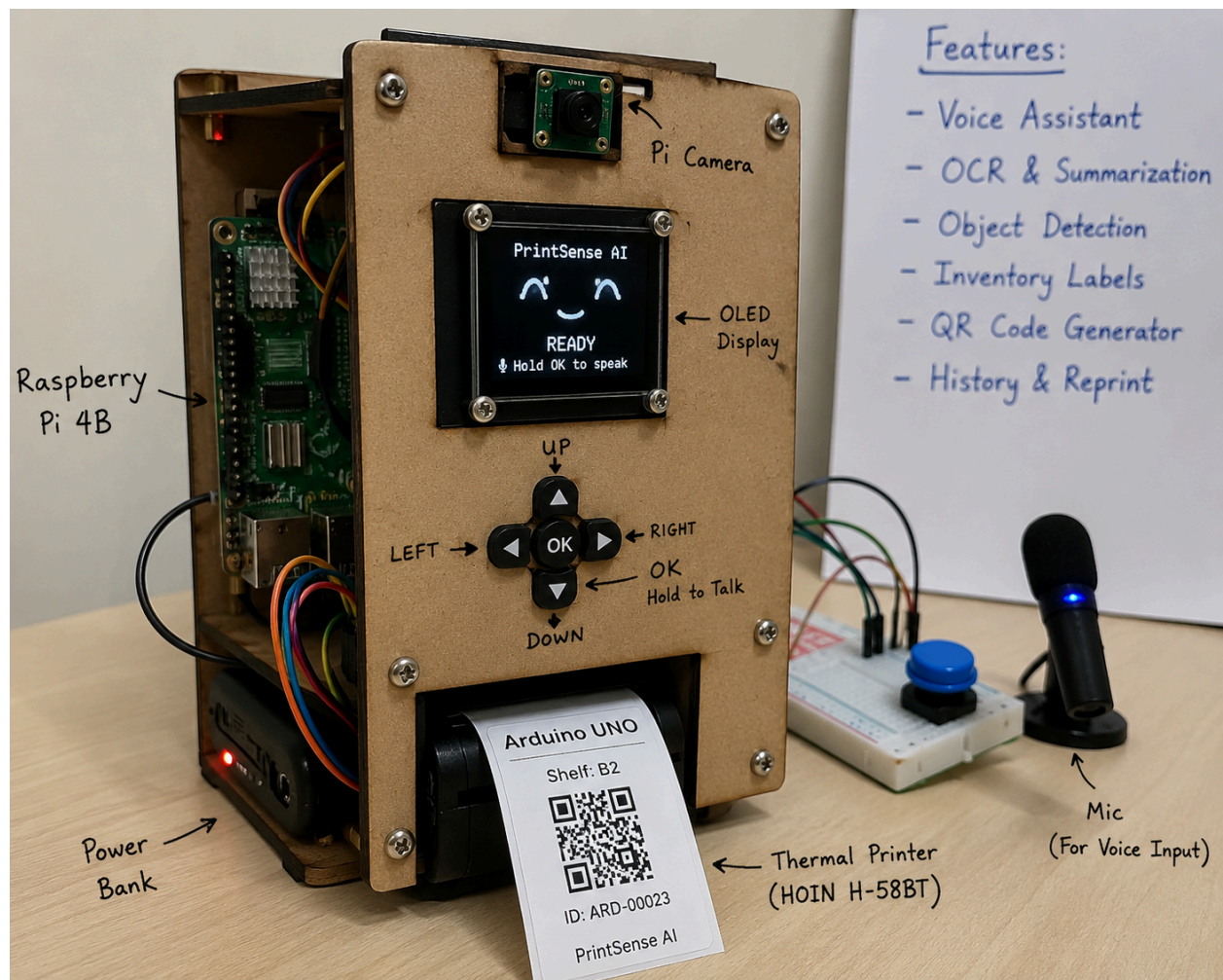
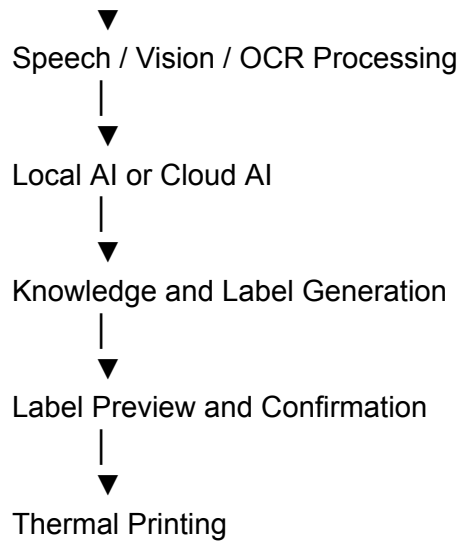
The proposed system will combine the following technologies:

- Voice-based interaction
- Camera-based object recognition
- Optical Character Recognition
- Local language models
- Cloud-based AI APIs
- QR-code generation
- Automatic label composition
- Thermal printing
- Physical button-based navigation
- OLED status display

The basic working process will be:

User Input





5. Main Functions of the System

5.1 Voice-Based Label Generation

The user speaks a command through the microphone. The system converts speech into text, understands the request, and generates the required label.

5.2 Object Recognition

The camera captures an object. An object-detection model identifies it and sends the detected information to the AI module.

5.3 OCR-Based Text Extraction

The system reads printed text from:

- Books
- Product packages
- Existing labels
- Documents
- Notes
- Formula sheets

The extracted information can then be summarized and printed.

5.4 QR-Code Generation

The system can generate QR codes containing:

- Website URLs
- Product information
- Inventory record links
- Study notes
- Contact details
- Wi-Fi credentials
- Locally hosted knowledge pages

5.5 Intelligent Thermal Printing

The system creates a complete monochrome label image containing:

- Heading
- Description
- QR code

- Barcode
- Date
- Category
- Quantity
- Shelf number
- Small icon or diagram

The image is converted into a printer-compatible bitmap and printed through the thermal printer.

6. Hardware Requirements

6.1 Final Hardware List

Sr. No.	Hardware Component	Quantity	Purpose	Estimated Cost
1	Raspberry Pi 4B, 4 GB RAM	1	Main processing unit and robot controller	Already available
2	Official Raspberry Pi Camera Module V2, 8 MP	1	Object detection, OCR, document scanning and image capture	₹1,700
3	HOIN H-58BT, 58 mm USB + Bluetooth thermal printer	1	Printing labels, QR codes, text, diagrams and stickers	₹2,100
4	2.42-inch 128×64 OLED display, SPI/I ² C	1	Robot face, menu, status messages and confirmations	₹1,200
5	USB plug-and-play microphone	1	Voice-command input	₹250
6	TV remote-style 5-way navigation switch	1	Up, down, left, right and OK menu navigation	₹150
7	Separate momentary push-to-talk button	1	Start and stop voice recording	₹100
8	WS2812B RGB LED ring or LED indicator module	1	Display listening, processing, printing and error states	₹160

9	Mini speaker or active buzzer	1	Voice response and notification sounds	₹300
10	32 GB Class 10 microSD card	1	Raspberry Pi OS, AI models, database and project software	₹500
11	Official 5.1 V, 3 A USB-C Raspberry Pi power supply	1	Stable power for Raspberry Pi	₹700
12	Raspberry Pi heatsink and cooling fan	1 set	Temperature control during AI-model inference	₹350
13	58 mm direct-thermal adhesive paper rolls	1 pack	Printable sticker media	₹350
14	Jumper wires, connectors, perfboard and resistors	1 set	Hardware wiring and circuit assembly	₹350
15	Prototype enclosure/body	1	Housing the Raspberry Pi, OLED, buttons and printer	₹1,000

₹9,210 approximately

8. Purpose of Each Major Component

Raspberry Pi 4B

The Raspberry Pi acts as the main brain of PrintSensei. It will:

- Run the operating system
- Control all connected devices
- Run lightweight offline AI models
- Manage cloud API requests
- Store print history
- Host the local dashboard
- Generate QR codes
- Prepare labels
- Control the printer

Camera Module

The camera will be used to:

- Capture physical objects
- Detect electronic components
- Scan printed documents
- Read product information
- Extract text using OCR
- Capture images for cloud vision APIs

OLED Display

The OLED will display:

READY
LISTENING
PROCESSING
OBJECT DETECTED
LABEL READY
PRINT?
PRINTING
DONE
ERROR

It can also show simple robot-face animations.

Navigation Controls

The TV remote-style navigation switch will provide:

UP

LEFT OK RIGHT

DOWN

The separate push-to-talk button will be held while the user speaks.

Thermal Printer

The printer will provide the physical output of the robot. It will print:

- Study cards
- Inventory tags
- Product labels
- QR stickers
- Barcodes
- Revision notes
- Object-information labels
- Small monochrome diagrams

LED Ring

Suggested colour states:

LED Colour	Robot State
White or green	Ready
Blue	Listening
Yellow	Processing
Purple	Generating label
Green animation	Printing
Red	Error

Related Literature

1. Large Language Models for Robotics: Opportunities, Challenges, and Perspectives

Year: 2025

DOI: 10.1016/j.jai.2024.12.003

Summary

This review paper examines how Large Language Models and multimodal Large Language Models can be integrated into robotic systems. It explains that LLMs provide strong natural-language understanding and reasoning abilities, allowing robots to interpret human instructions and convert them into structured task plans.

The paper discusses the use of LLMs in robotic task planning, reasoning and manipulation. However, it notes that text-only language models cannot directly understand the visual and physical conditions of a robot's environment. This limitation makes it difficult for them to complete embodied tasks that depend on cameras and other sensors.

To address this problem, the authors present a framework using the multimodal GPT-4V model. The framework combines natural-language instructions with visual observations from the robot to improve embodied task planning. The reported evaluation shows that integrating language understanding with visual perception can improve robotic task performance and support better interaction between humans, robots and their environments. ([SciOpen](#))

Concepts Related to PrintSensei

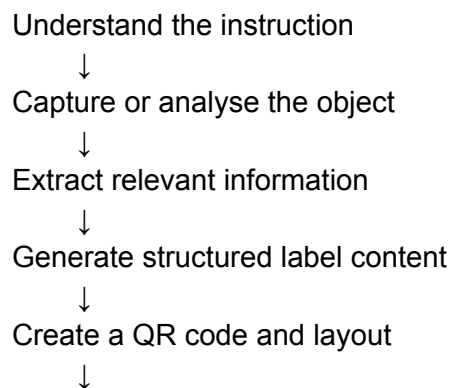
The following concepts from this paper are directly related to PrintSensei:

- Natural-language command understanding
- LLM-based reasoning and task planning
- Integration of language and camera-based perception
- Multimodal AI for robotic applications
- Human–robot interaction
- Connecting AI-generated decisions with physical robot actions

In PrintSensei, the user provides a voice or text instruction, such as:

“Create an inventory label for this object.”

The system then performs a sequence of operations:



Send the result to the thermal printer

Therefore, PrintSensei applies the paper's concept of using language models as a reasoning and coordination layer. The main difference is that the final physical action in PrintSensei is the production of an intelligent printed label.

2. NaviBlind: A Multimodal AI Assistant for Visually Impaired Users to Identify Product Information from Images and Speech

Year: 2025

DOI: 10.22144/ctujoisd.2025.057

Summary

NaviBlind is a multimodal AI-powered assistive system developed to help visually impaired users obtain information about physical products. The system allows a user to upload or capture a product image and provide a spoken question about the product, such as its name, colour, type or expiry date.

The system uses the Gemini Flash vision model to understand product images and PhoWhisper to recognise Vietnamese speech commands. It combines the visual input and the user's spoken request to generate a concise and structured description of the required product information. The resulting text is then converted into speech for the user.

The system also includes mechanisms for identifying uncertain or potentially hallucinated responses, particularly when the supplied product image has poor quality. NaviBlind is implemented as a user-friendly web application and demonstrates how image understanding, speech recognition and vision-language models can be combined to provide real-time product assistance. ([CTUJS](#))

Concepts Related to PrintSensei

The following NaviBlind concepts are closely related to PrintSensei:

- Image and speech as combined inputs
- Product and object identification
- Vision-language model integration
- Natural voice-based interaction
- Structured information extraction

- Real-time multimodal processing
- Hallucination and uncertainty handling
- User-friendly web interface

The operating process of NaviBlind is:

Product Image + Voice Question



Multimodal AI Processing



Structured Product Information



Speech Output

The proposed PrintSensei follows a similar processing approach:

Object/Image + Voice Instruction



Vision, OCR and AI Processing



Structured Knowledge Generation



Label Design + QR Code Generation



Physical Thermal Printing

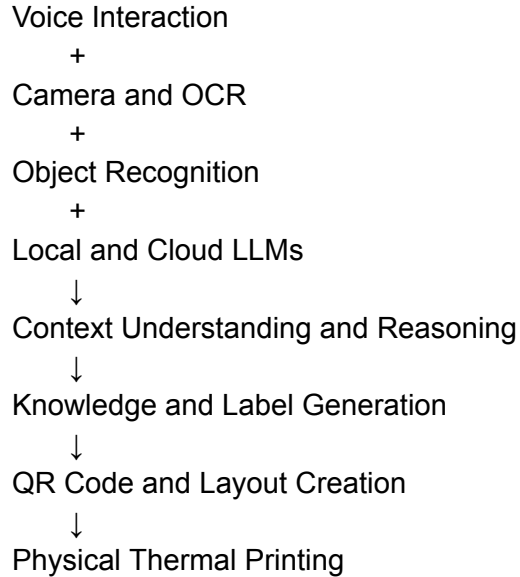
The major difference is the output method. NaviBlind provides product information through speech, whereas PrintSensei transforms the extracted or generated information into a physical adhesive label. The printed label may include text, inventory information, a QR code, a barcode or a brief knowledge summary.

Combined Relationship with PrintSensei

Both papers demonstrate the importance of combining language understanding, visual perception and AI reasoning in intelligent interactive systems.

The first paper provides the theoretical foundation for using an LLM as the reasoning and task-planning component of a robot. The second paper provides an application-oriented example of combining images, speech recognition and a multimodal vision model to extract structured information about physical products.

PrintSensei combines these concepts in one system:



Thus, PrintSensei extends existing multimodal AI-assistant concepts by introducing **automatic physical knowledge and label generation** as the robot's final action.
